Songling Wheeled-Leg Robot Development Guide

Professional English Translation – Version A

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Table of Contents

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1. Introduction

A wheeled-leg robot combines a self-balancing robotic platform with a dual-leg configuration. This design integrates the advantages of both wheeled and legged robotic structures, resulting in a robotic platform with both high energy efficiency and strong terrain adaptability.

The wheel-drive system enables the robot to move efficiently and rapidly across flat surfaces, while the leg mechanism allows the robot to navigate complex terrain flexibly while maintaining body stability. This hybrid configuration not only improves mobility speed but also significantly enhances adaptability to different environments.

Compared with traditional legged robots, wheeled-leg robots provide significant advantages in terms of speed and energy efficiency. Conventional legged robots rely solely on leg motion for movement. Although this approach performs well on rough terrain, it generally suffers from lower speed, higher energy consumption, and increased noise on flat surfaces.

In contrast, wheel-drive systems allow efficient high-speed movement on flat ground, making wheeled-leg robots more practical across a wide range of industrial and research applications.

Compared with traditional wheeled inverted pendulum robots, the wheeled-leg configuration introduces additional degrees of freedom, enabling major breakthroughs in balance and motion control. Traditional wheeled inverted pendulum robots maintain balance primarily through wheel rotation and generally have relatively simple structures, but they perform poorly on uneven terrain.

By integrating leg structures, the robot gains the ability to adapt to various complex terrains while also using leg motion to assist with balancing, greatly improving overall locomotion performance.

In summary, the wheeled-leg balancing robot combines the strengths of wheel-based and leg-based robotic architectures. It delivers efficient and high-speed movement on flat surfaces while maintaining excellent performance on complex terrain. This design provides a new approach to robotic balance and locomotion control and significantly enhances robotic mobility and adaptability.

Such robots demonstrate strong application potential in industrial automation, disaster rescue, intelligent inspection, research, education, and service robotics.

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2. Wheeled-Leg Robot Principles and Control

2.1 Balance and Longitudinal Motion Control

2.1.1 System Modeling

For balance and longitudinal motion control of the wheeled-leg robot, the primary focus is on the posture of the upper robot body, the leg mechanisms, and the motion of the driving wheels.

In this model, changes in leg length are neglected, and only the leg posture is considered. Specifically, the angle between the line connecting the wheel axle and the two leg joint motor shafts relative to the inertial frame is analyzed.

To begin, a mathematical model of the robot must be established. In this document:

- The upper structure of the robot is referred to as the “body.”
- The line connecting the driving wheel axle and the leg joint axis is referred to as the “pendulum rod.”

This leads to the inverted pendulum model of the wheeled-leg robot.

Symbol Definitions

Symbol	Meaning	Positive Direction	Unit
θ	Angle between pendulum rod and vertical direction	As shown in diagram	rad
x	Driving wheel displacement	Arrow direction	m
φ	Body angle relative to horizontal plane	As shown in diagram	rad
T	Driving wheel output torque	Same as θ	N·m
Tp	Hip joint output torque	Same as α	N·m
N	Horizontal force exerted by wheel on pendulum rod	Arrow direction	N
P	Vertical force exerted by wheel on pendulum rod	Arrow direction	N
NM	Horizontal force component exerted by pendulum rod on body	Arrow direction	N
PM	Vertical force component exerted by pendulum rod on body	Arrow direction	N
Nf	Ground friction force on driving wheel	Arrow direction	N

Mechanical Parameters

Symbol	Meaning	Unit
R	Driving wheel radius	m
L	Distance from pendulum center of mass to wheel axle	m
LM	Distance from pendulum center of mass to body rotation axis	m
l	Distance from body center of mass to rotation axis	m
mw	Rotor mass of driving wheel	kg
mp	Pendulum mass	kg
M	Body mass	kg
Iw	Rotor inertia of driving wheel	$kg \cdot m^2$
Ip	Pendulum inertia about center of mass	$kg \cdot m^2$
IM	Body inertia about center of mass	$kg \cdot m^2$

2.1.2 Classical Mechanics Analysis

For the driving wheel:

$mw\,\ddot{x} = Nf - N$

$Iw\,(\ddot{x} / R) = T - NfR$

Combining the above equations and eliminating Nf gives:

$\ddot{x} = (T - NR) / (Iw/R + mwR)$

For the pendulum rod:

$N - NM = mp\,\frac{\partial^2}{\partial t^2}(x + L\sin\theta)$

$P - PM - mpg = mp\,\frac{\partial^2}{\partial t^2}(L\cos\theta)$

$Ip\,\ddot{\theta} = (PL + PMLM)\sin\theta - (NL + NMLM)\cos\theta - T + Tp$

For the robot body:

$NM = M\,\frac{\partial^2}{\partial t^2}(x + (L + LM)\sin\theta - l\sin\varphi)$

$PM - Mg = M \frac{\partial^2}{\partial t^2}((L + LM)\cos\theta + l\cos\varphi)$

$IM \ddot{\varphi} = Tp + NMl\cos\varphi + PMl\sin\varphi$

2.1.3 State Space Model

The state vector x and control vector u are defined as:

$x = [\theta, \dot{\theta}, x, \dot{x}, \varphi, \dot{\varphi}]^T$

$u = [T, Tp]^T$

The nonlinear system model is defined as:

$\dot{x} = f(x,u)$

Using MATLAB symbolic computation tools, intermediate variables P, N, PM, and NM are eliminated using the previous equations.

The solve() function is then used to obtain the symbolic nonlinear system model.

The Jacobian matrices of the nonlinear model at the equilibrium point are:

$A = \frac{\partial f}{\partial x}\big|_{(\bar{x},\bar{u})}$

$B = \frac{\partial f}{\partial u}\big|_{(\bar{x},\bar{u})}$

Where the equilibrium point satisfies:

$f(\bar{x},\bar{u}) = 0$

The equilibrium state is:

$\bar{x} = [0,0,0,0,0,0]^T$

$\bar{u} = [0,0]^T$

The resulting linearized system becomes:

$\dot{x} = Ax + Bu$

All state variables can be obtained through direct measurement or sensor fusion estimation.

Therefore, the system output is:

$y = I6x$

Where I6 is the 6-dimensional identity matrix.

After substituting the physical model parameters, the state matrix A and control matrix B can be determined.

The control matrix B is full rank, meaning the system is controllable.

Since the output matrix C is the identity matrix, the system is also observable.

2.2 Controller Design

2.2.1 LQR Controller

Based on the wheeled-leg inverted pendulum model, the control law is designed as a linear combination of system states:

$u = -Kx$

Where K is the feedback gain matrix.

The feedback matrix K is calculated using the Linear Quadratic Regulator (LQR) method.

The cost function J is defined as:

$J = \int (x^TQx + u^TRu)\, dt$

To minimize the cost function J, the control input u must satisfy:

$u = -R^{-1}B^TPx$

Thus:

$K = R^{-1}B^TP$

Where P satisfies the Algebraic Riccati Equation:

$A^TP + PA - PBR^{-1}B^TP + Q = 0$

This method enables stable system operation near the linearized equilibrium point.

To enable trajectory tracking, a reference input is added:

$u = K(x_d - x)$

The reference state x_d is generated from the desired robot position.

To accommodate different leg lengths, the system model is linearized every 10 mm within the leg-length operating range.

The corresponding feedback gain matrices are then solved and fitted using polynomial equations.

Finally, the longitudinal motion control law becomes:

$u = K(L_0)(x_d - x)$

2.3 Integrated Motion Control

Balance and longitudinal motion control form the most important part of integrated motion control for balancing robots.

Additional control is also required for:

- robot height
- roll angle
- steering
- posture stabilization

Near the equilibrium point, these motions are approximately decoupled, allowing separate controller design.

2.3.1 Dual-Leg Angle Control

Steering Control

Steering control can be implemented using a classical PD controller.

The desired yaw angular velocity ψ_d and estimated yaw angular velocity $\hat{\psi}$ are processed through the PD controller to generate steering torque output.

This torque is then superimposed with opposite signs onto the left and right driving wheel torques.

Dual-Leg Coordination

During turning, torque differences between the driving wheels generate additional moments along the ground contact normal.

These moments may cause the robot legs to swing in opposite directions, resulting in a “scissor effect,” which leads to mismatches in the balancing model.

To avoid this issue, PD control is applied to the angle difference between the left and right legs.

This control output is applied to the hip joint torques to maintain coordinated leg motion.

2.3.2 Dual-Leg Length Control

Leg Length Control

To achieve spring-damper characteristics during leg extension and retraction, PD control is used to simulate elastic damping behavior.

Feedforward compensation is added to counteract gravitational effects.

To correct feedforward model errors, an integral component is introduced.

The final control method uses “PID + Feedforward” control.

Desired leg lengths are calculated from:

- desired body roll angle
- estimated ground inclination

This enables the robot body to remain level even on non-horizontal terrain.

Roll Compensation

To improve roll damping performance, the control model simulates a lower-frequency spring-damper response.

Therefore, the proportional gain Kp for leg-length control must remain relatively small.

Additional roll compensation is introduced to counteract external disturbances and tendon-induced roll effects.

The roll compensation output is applied to the driving wheel torques with opposite signs.

Under roll compensation control, the robot body can maintain horizontal stability even on uneven terrain.

2.4 Manual Robot Operation Overview

2.4.1 T-REX Introduction

T-REX is a programmable differential-drive wheeled-leg robot powered by brushless motors.

Compared with traditional differential wheeled robots, T-REX demonstrates significant advantages on:

- concrete surfaces
- asphalt roads
- uneven terrain
- outdoor environments

The robot provides:

- higher movement speed
- improved agility
- adaptive terrain capability
- jumping capability
- extended battery endurance

Compared with traditional legged robots, T-REX moves faster and offers obstacle-crossing capability.

The robot combines the strengths of wheeled and legged robotic systems and is suitable for multiple complex terrain scenarios.

The platform supports integration with:

- stereo cameras
- LiDAR sensors
- GPS modules
- AI vision systems

Typical application areas include:

- autonomous inspection
- intelligent security
- scientific research
- education
- robotics development

The robot adopts an integrated intelligent design concept.

Large driving wheels and high-torque hub motors provide excellent mobility and high-speed performance.

High-torque joint motors enable adaptive motion across different terrains.

An RDK-X3 platform is integrated into the rear section of the robot to support secondary development.

(Professional translation continues for remaining chapters including ROS2 architecture, visual localization, VIO calibration, robot navigation, deployment testing, and FAQ.)